Higher National Diploma in Engineering (Civil/Electrical/Mechanical), Higher National Diploma In Building Service Engineering, and Higher National Diploma in Quantity Surveying

Second Year, First Semester Examination-2019

IT2102-Information Technology II A

Answer only 4 questions **Time: 2 hours**

Question 1

i. What is a computer programming language? give two examples for that (03 Marks)

ii. What is the purpose of a compiler/interpreter? (02 Marks)

iii. Visual Basic is an event driven programming language. Write any five events. (05 Marks)

iv. Following program is written to change the background of the form. Identify the:

- **a.** Names of the controls (02 Marks)
- **b.** Name of the event (01 Mark)
- **c.** Property of the control (01 Mark)
- **d.** Value assigned to the property (01 Mark)

```
Private Sub cmdBColor_Click()

frmMain.BackColor = vbRed

End Sub
```

v. Evaluate 5 Mod 2 * 4^2 * 5 - 2 + 7 \ 4 + 6 / 2 (05 Marks)

vi. Write examples for:

a. Two arithmetic operators (02 Marks)
b. Two relational operators (02 Marks)
c. One logical/bitwise operator (01 Mark)

Question 2

i. What is the difference between a **variable** and a **constant**? (06 Marks)

ii. Identify valid and invalid identifiers (variable names)

a. intMarks **b.** sub **c.** Boolean **d.** stu Age **e.** @Name

f. 9students (1 x 06 Marks)

iii. Declare variables/ constants to store values in the table (05 Marks)

Situation	**Value**
a. Saman's IT marks	75
b. Value of π	3.14
c. Price of a bun	Rs. 30.50
d. The average marks of Maths	46.89748264884
e. Student's registration number	GALEE2018F0001

iv. Briefly explain the scope of the variables **x** and **y** in bellow program (04 Marks)

```
Dim x As Integer
Private Sub Command1_Click()
Dim y As Integer
End Sub
```

v. Write the output of the following two programs (2x04 Marks)

a.

```
Dim x, y, z As Integer
x = 4
y = 5
z = x
x = y
y = z
Print x & y
```

b.

```
Dim x, y, z As Integer
x = 4
y = x ^ 2
z = y \ 3
Print x & y & z
```

Question 3

i. What do you mean by the **control structures**? Give three examples for that (05 Marks)

ii. Write the output of the following programs (2x03 Marks)

a.

```
Dim x As Integer
x = 0
Do While (x < 5)
Print x
x = x + 1
Loop
```

b.

```
Dim y As Integer
y = 10
Do
Print y
y = y + 1
Loop Until (y > 0)
```

c.

```
Dim z, x As Integer
z = 10
x = 1
Do
Print z
z = z - x
x = x + 2
Loop Until (z < 0)
```

iii. Following program shows health condition according to the given criteria. A fasting blood sugar level less than 100 mg/dL is **normal**.A fasting blood sugar level from 100 to 125 mg/dL is considered **prediabetes**. If it is 126 or higher you have **diabetes**.

a. Design (draw) an interface to display health condition according to the entered blood sugar level. You need to display output on a **label** and name all controls used (04 Marks)

b. Write the code for above interface (04 Marks)

iv. Using a loop, write a program to display sum of bellow series (1, 4, 7, 10, 13, 16, 19, 22) (04 Marks)

v. Write a program to display the bellow pattern (02 Marks)

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Question 4

i. What is an **array**? (03 Marks)

ii. Using an array:

a. write a program to store following marks of Nimal, got at the final exam (40, 57, 34, 78, 87, 32, 20, 67, 56, 24, 73, 26) (04 Marks)

b. using a loop, print all marks which are greater than 40 (in part a.) (04 Marks)

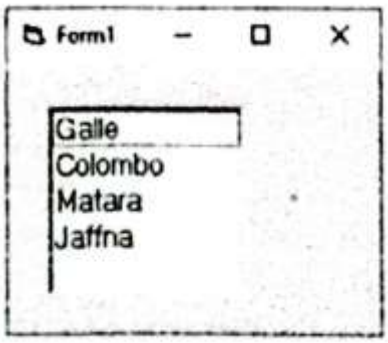
c. using a loop, print the average marks of Nimal (04 Marks)

iii. The following program is written to print **SLIATE** but due to an error of the loop it displays **ETAILS**. Correct that loop to print the expected output (04 Marks).

```
Dim x(5) As String
x(0) = "E"
x(1) = "T"
x(2) = "A"
x(3) = "I"
x(4) = "L"
x(5) = "S"

Dim i As Integer
For i = 0 To 5
Print x(i);
Next
End Sub
```

iv. Create an array to store following cities **Galle, Colombo, Matara and Jaffna**. When the form is loading, display all city names in that array on a list box as shown in the below interface (06 Marks).



Question 5

i. Compare and contrast **function procedures** and **event/sub procedures** (03 Marks)

ii. Write the syntax of function and event procedures (04 Marks)

iii. Write three advantages of function/event procedures (03 Marks)

iv. Write a user defined function (UDF) to calculate tax amount (up to 2000 cc) that you need to be paid when importing a petrol car. Engine capacity and appropriate tax display in the bellow table:

Engine Capacity x (cm^3)	Tax (For one cubic centimeter) Rs.
X <=1000	1750.00
1000< x <=1300	2750.00
1300< x <=1500	3250.00
1500< x <=1600	4000.00
1600< x <=1800	5000.00
1800< x <=2000	6000.00

In the program you need to enter engine capacity to the textbox (txtCapacity) and when you click on the command button (cmdCal) it will call the function (calTax) and display the tax amount to be paid on the label (lblOut)

a. Write a function to return tax amount relevant to the given engine capacity (05 Marks)
b. Call above function to display the output. (03 Marks)

v. Write an event/sub procedure to display the cube power (x^3) of any given whole number on a **massage box**. Then, execute that event/sub procedure by passing a value to it. (05 Marks)